Renormalised One-Loop Quantum Defect Boundary Theory and Universal KP-Triangle Holonomy Proofs for the Good Boussinesq Equation

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4 November 2026

Abstract

We carry out ranked next step #4 of the revised Phase 2 master report (`phase2/report.tex`): constructing the renormalised one-loop quantum boundary theory for the Corrigan–Zambon (CZ) type-I integrable defect in the good Boussinesq equation and proving the canonical-charge and absorbed-momentum KP-triangle holonomy statements across *all* nondegenerate KP triangles, incorporating all six corrections from the independent audit `phase2/audit_defq.tex` (`phase2/errata.tex` entry N07).

1. **Renormalised one-loop boundary theory and uniqueness of counterterms (c) (`phase2/scripts/defb_boundary_renorm.py`):** With the corrected plus sign in the quadratic energy trace, $\langle E_{D,2}\rangle_q + \langle E_{D,2}\rangle_{-q} = \langle K\rangle_q + \langle K\rangle_{-q} = +\frac{\omega(q)}{4\pi}\big(\theta'_D(q)+\theta'_D(-q)\big) = -\frac{\sqrt{3}\eta\sigma_{\rm vac}}{4\pi} + O(q^{-2})$, we derive *exact real elementary closed forms* for the finite one-loop Casimir momentum $\Delta P_D^{\rm Cas}(\sigma_{\rm vac},\eta)$, the density-subtracted Casimir energy $\mathcal{E}_D^{\rm sub}(\sigma_{\rm vac},\eta)$, and the mode-sum renormalised Casimir energy $\Delta E_D^{\rm ren}(\sigma_{\rm vac},\eta)$ (verified against the two-root dipole representation to 1.36×10^{-71} and compact-interval rational quadrature to 4.81×10^{-72}). Within the integrable CZ parameter family (s,ρ_0,τ_0,h_0), an exact *envelope theorem* on static vacua ($\tau'(\sigma_i)=h'(\sigma_i)=0$) proves that $O(\hbar)$ shifts $\delta\sigma_i$ of the vacuum roots and δs_1 of the lower integration limit contribute *zero* at $O(\hbar)$ to $E_D^{\rm vac}(q_i)$ and $P_D^{\rm vac}(q_i)$. Imposing (i) preservation of the $w_\varphi=0$ vacuum ($\delta\rho_{0,1}=0$), (ii) minimal ground-state subtraction at the reference stable vacuum ($\delta\tau_{0,1}=\delta h_{0,1}=0$), and (iii) the physical zero-momentum transmission slope (scattering length) $a_D(s,\eta) = \theta_D^{\rm phys\prime}(0^+;s,\eta) = -\frac{2\sqrt{3}\eta}{s+1}$ (since $z_{\rm ph}(0;s,\eta)=+1$ is topologically pinned) uniquely fixes $(\delta s_1,\delta\rho_{0,1},\delta\tau_{0,1},\delta h_{0,1}) = (0,0,0,0)$. Including the bulk vacuum-elasticity counterterm $\Delta\mathcal{H}_{\rm bulk} = \hbar\lambda_g u^2$ of `phase2/onebody_scheme.tex` shifts the directed triangle sum of mode-sum soliton energies by $2\lambda_g\eta(q_1-q_2)(q_2-q_3)(q_3-q_1)\neq0$, uniquely selecting the continuum zero-sound-shift scheme $\lambda_g=0$ for mode-sum edge balance.
2. **Recomputed $O(\hbar^0)$ transmission phase and stable-vacuum absorption analysis (c) (`phase2/scripts/defb_boundary_renorm.py`):** Using the audited convention $s=\sigma_{\rm vac}=-2\sqrt{3}q_D$ for scattering and $c=\eta\sigma_{\rm vac}=-2\sqrt{3}\eta q_D$ for the decaying-mode spectrum, we verify genuine $z_t=0$ time-dependent absorption conservation to 2.26×10^{-72} (replacing the old script's spurious input $s=+1.5$ with $z_t=\sqrt{21}-3\neq0$). Because $f'(q)=12q^2-1>0$ on the stable domain $|q|>\frac{1}{2\sqrt{3}}$, at most one root of the KP cubic $4q^3-q=$ const is linearly stable ($c\le-1$) for a fixed orientation η; thus the only absorption/emission balance whose external defect states are linearly stable vacua is the composite closed-cycle process $D_\eta(q_{\rm st})+S_1+S_2 \leftrightarrow D_\eta(q_{\rm st})+\bar{S}_3$ returning to the unique stable ground state $q_{\rm st}$.
3. **Universal KP-triangle holonomy theorem (c) (`phase2/scripts/defb_boundary_holonomy.py`):** Reducing $\Delta P_{\rm can}(k,v)$ and $\Delta E_{\rm can}(k)$ to elementary functions of $k=\sin\alpha$ and $v=\epsilon\cos\alpha$ and parametrising all nondegenerate KP triangles on the fundamental domain $\vartheta\in(0,\frac{\pi}{3})$, we prove by exact endpoint asymptotics, exact symmetry-point derivatives at $\vartheta=\frac{\pi}{6}$ (where $\mathcal{C}_P(\frac{\pi}{6})=0$, $\mathcal{C}_E(\frac{\pi}{6}) = -\frac{3\sqrt{3}}{4\pi}\ln(2+\sqrt{3})\,\eta\neq0$, and

$\mathcal{C}'_P(\frac{\pi}{6}) = \frac{3\sqrt{3}\ln(2+\sqrt{3})-9}{\pi}\,\eta \neq 0$), and rigorous `mpmath.iv` interval arithmetic across 2000 subintervals that $\eta\,\mathcal{C}_E(\vartheta) < 0$ for *every* $\vartheta \in (0, \frac{\pi}{3})$ and $\mathcal{C}_P(\vartheta) = 0$ on $(0, \frac{\pi}{3})$ *if and only if* $\vartheta = \frac{\pi}{6}$. Thus $(\mathcal{C}_E, \mathcal{C}_P) \neq (0,0)$ for *every* nondegenerate KP triangle, and $\mathcal{C}_P(\vartheta)$ is strictly invariant under the bulk elasticity counterterm λ_g. We likewise prove universally that the absorbed-soliton momentum remainder $\mathcal{R}_{\mathrm{abs}}$ has a strictly negative branch-point coefficient $w_1(\vartheta) - w_2(\vartheta) < 0$ on all $\vartheta \in (0, \frac{\pi}{3})$, and derive the exact pole-free identity $\partial_{q_i}\delta_{D,\mathrm{cl}}(3; q_i, \eta) = -\frac{2\sqrt{3}}{3}(12q_i^2 - 1)\epsilon_3 \ln|z_t(3; q_i, \eta)| + 4\epsilon_3\eta k_3(v_3 + 2\sqrt{3}q_i)$.

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1 Conventions, dictionary, and incorporation of the `audit_defq` verdicts

1.1 Fixed project conventions and the two defect parameters s and c

Throughout this report we work on the **good** Boussinesq branch in the fixed conventions of `AGENTS.md`:

$$u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}, \qquad u = \varphi_x = 2(\log\tau)_{xx}, \tag{1}$$

$$\mathcal{L} = \frac{1}{2}\varphi_t^2 - \frac{1}{2}\varphi_x^2 + \varphi_x^3 - \frac{1}{2}\varphi_{xx}^2, \qquad \Omega = \int \mathrm{d}\pi \wedge \mathrm{d}\varphi, \quad \pi = \varphi_t. \tag{2}$$

A classical one-soliton state $S = (\alpha, \epsilon)$ with $\alpha \in (0, \frac{\pi}{2}) \setminus \{\frac{\pi}{3}\}$ and $\epsilon \in \{+1, -1\}$ has amplitude parameter $k = \sin\alpha$, velocity $v = \epsilon\cos\alpha$, frequency $\omega_{\mathrm{sol}} = kv$, spectral parameter $\ell = -\epsilon\cos 3\alpha$, KP roots $p_a = \frac{1}{2}(k + v/\sqrt{3}) > p_b = \frac{1}{2}(v/\sqrt{3} - k)$ satisfying $p_a^2 + p_a p_b + p_b^2 = \frac{1}{4}$ and $\ell = 3\sqrt{3}(4p_{a/b}^3 - p_{a/b})$, and conserved classical bulk charges **(a)**/**(c)**:

$$M = 2k, \qquad \Pi = -Mv = -2\epsilon k\cos\alpha, \qquad P = \frac{2}{3}\epsilon k^3\cos\alpha, \qquad E = \frac{2}{15}k^3(5 - 4k^2). \tag{3}$$

In the integrable type-I defect construction of Corrigan and Zambon (CZ) [1] **(a)** (with dimensional constants $c_{\rm CZ} = \lambda = \kappa = 1$), the bulk potential $u_{\rm CZ}$ satisfies $\mathcal{L}_{\rm CZ} = \frac{1}{2}u_{{\rm CZ},t}^2 - \frac{1}{2}u_{{\rm CZ},x}^2 - \frac{1}{3}u_{{\rm CZ},x}^3 - \frac{1}{2}u_{{\rm CZ},xx}^2$, giving the exact project dictionary **(c)** (`phase2/scripts/defb_boundary_renorm.py`):

$$u_{\rm CZ} = -3\varphi_L\ (x<0), \qquad v_{\rm CZ} = -3\varphi_R\ (x>0), \qquad \mathcal{L}_{\rm CZ} = 9\,\mathcal{L}, \qquad D_{\rm proj} = \frac{1}{9}D_{\rm CZ}, \qquad (E_{\rm CZ}, P_{\rm CZ}) = 9\,(E,P). \tag{4}$$

Following Section 2 of the independent audit `phase2/audit_defq.tex` **(c)**, we distinguish explicitly between the **scattering parameter** s and the **spectral stability parameter** c:

$$\boxed{s = \sigma_{\rm vac} = -2\sqrt{3}\,q_D, \qquad q_D = -\frac{s}{2\sqrt{3}}, \qquad c := \eta\,\sigma_{\rm vac} = \eta\,s = -2\sqrt{3}\,\eta\,q_D, \qquad \eta \in \{+1,-1\}.} \tag{5}$$

Here $s = \sigma_{\rm vac}$ enters the classical and linearised scattering transmission factors directly,

$$z_t(\alpha,\epsilon;q_D,\eta) = \left(\frac{v+s-\eta\sqrt{3}\,k}{v+s+\eta\sqrt{3}\,k}\right) = \left(\frac{p_b-q_D}{p_a-q_D}\right)^{\eta}, \tag{6}$$

$$z_{\rm ph}(q;s,\eta) = \left(\frac{\mathrm{sgn}(q)\sqrt{1+q^2}+s-\mathrm{i}\sqrt{3}\,q}{\mathrm{sgn}(q)\sqrt{1+q^2}+s+\mathrm{i}\sqrt{3}\,q}\right)^{\eta}, \tag{7}$$

whereas the linearised decaying-mode determinant on a constant defect vacuum $\sigma_{\rm vac}$ depends on the product $c = \eta\sigma_{\rm vac}$:

$$\frac{\det M_{4\times4}(g;\sigma_{\rm vac},\eta)}{(\kappa_1-\kappa_2)^2} = -\frac{2}{3}\,S^2(S^2-1)\,P_-(S;c)\,P_+(S;c), \qquad P_\mp(S;c) = S^2 - \frac{\sqrt{3}\mp1}{2}cS + \frac{c^2-2\mp\sqrt{3}}{2}, \tag{8}$$

with $S = \kappa_1+\kappa_2 = \sqrt{1+2g} > 1$ and growth rate $g = \frac{1}{2}(S^2-1) > 0$ **(c)**.

1.2 Verification of the six corrections from `audit_defq.tex`

Before constructing the renormalised boundary theory, our first script (`phase2/scripts/defb_boundary_renorm.py`, Section 1) independently verifies all six corrections identified by `phase2/audit_defq.tex` (`phase2/errata.tex` entry N07):

Proposition 1.1 (Verification of the six `audit_defq` corrections, **(c)**, `phase2/scripts/defb_boundary_reno`

True $z_t = 0$ ***soliton absorption vs spurious extra*** η***:*** *At* $q_1 = \frac{\sqrt{3}}{4}$, $q_2 = \frac{\sqrt{7}-\sqrt{3}}{8}$, *and* $\eta = -1$, *passing* $s_{\rm wrong} = -2\sqrt{3}q_1\eta = +\frac{3}{2}$ *into a function expecting bare* $s = -2\sqrt{3}q_1 = -\frac{3}{2}$ *yields regular transmission with the exact algebraic factor*

$$z_{\rm wrong} = \frac{12}{3+\sqrt{21}} = \sqrt{21}-3 = 1.582575694955840006588047\ldots \neq 0. \tag{9}$$

With the correct bare parameter $s = \eta\sqrt{3}k - v = -2\sqrt{3}q_1 = -\frac{3}{2}$ *(for* $\eta=-1$*) or* $s = -2\sqrt{3}q_2$ *(for* $\eta=+1$*), transmission vanishes identically (*$z_t = 0$*) and direct half-line quadrature of* $(E_{\rm tot}, P_{\rm tot}, \Pi_{\rm tot}, M_{\rm tot})(t)$ *at* $t \in \{-4, 0, 3.5\}$ *is conserved to* 2.26×10^{-72}.

2. ***Plus sign in the quadratic energy trace and*** $\langle J_{\rm bulk,2}\rangle_q = 0$***:*** *For the quadratic defect Lagrangian* $D_2 = a_tF_1 + b_tG_1 - E_{D,2}$ *with* $K := a_tF_1 + b_tG_1$ *and* $E_{D,2} = -A\sigma_0(a_x^2 + a_xb_x + b_x^2) - \frac{1-\sigma_0^2}{2}(a-b)(a_x+b_x) + \frac{A\sigma_0(1-\sigma_0^2)}{4}(a-b)^2$ *(*$A = \eta/\sqrt{3}$*), the bulk integration-by-parts boundary term* $J_{\rm bulk,2}$ *has* $\langle J_{\rm bulk,2}\rangle_q = 0$ *on reflectionless plane waves, and the symmetric* $\pm q$ *trace obeys the exact pointwise identity with a* ***plus*** *sign:*

$$\boxed{\langle E_{D,2}\rangle_q + \langle E_{D,2}\rangle_{-q} = \langle K\rangle_q + \langle K\rangle_{-q} = +\frac{\omega(q)}{4\pi}\Big(\theta_D'(q;s,\eta) + \theta_D'(-q;s,\eta)\Big) = -\frac{\sqrt{3}\,\eta\,\sigma_0}{\pi}\,\frac{q\sqrt{1+q^2}(4q^2}{16q^4+4(\sigma_0^2+2)q^2}} \tag{10}$$

(In `defq_oneloop_renorm.py`, a sign error in the printed string on line 350 produced a false minus sign in Eq. (27) of `defq_charges_fusion.tex`, while a typo replacing $a_x^2 + a_x b_x + b_x^2 \mapsto q^2(2 + \mathrm{Re}\, z)$ by $q^2(1 + 2\,\mathrm{Re}\, z)$ on line 293 produced a spurious mismatch between $\langle E_{D,2}\rangle_{\mathrm{sym}}$ and $\langle K\rangle_{\mathrm{sym}}$.)

3. ***Real-mode vacuum covariance:*** *A real cosine mode $\frac{1}{\sqrt{\pi\omega}}\,\mathrm{Re}(\mathrm{e}^{\mathrm{i}(qx-\omega t)})$ has ordinary time average $\frac{1}{2\pi\omega}$; obtaining the standard vacuum contraction $\langle a^2\rangle_q = \frac{1}{4\pi\omega(q)}$ used in (10) requires either mode amplitude $\frac{1}{\sqrt{2\pi\omega(q)}}$ or an explicit zero-point weight of $\frac{1}{2}$ upon time averaging.*

4. ***Coincident signed-wavenumber limits at $\pm\infty$:*** *Because both $\mathrm{sgn}(q)\sqrt{1+q^2}$ and $-\mathrm{i}\sqrt{3}q$ change sign under $q \mapsto -q$, the phonon transmission factor (7) has the* same *limit at both infinities:*

$$\lim_{q\to+\infty} z_{\mathrm{ph}}(q;s,\eta) = \lim_{q\to-\infty} z_{\mathrm{ph}}(q;s,\eta) = \mathrm{e}^{-2\pi\mathrm{i}\eta/3}. \tag{11}$$

5. ***Exact growth rate at the branch-switch threshold $c = \sqrt{3}$:*** *At $c = \eta\sigma_{\mathrm{vac}} = \sqrt{3}$, the marginal root has $g_2(\sqrt{3}) = 0$ ($S = 1$) and the remaining strictly positive admissible growth rate from (8) is*

$$g_1(\sqrt{3}) = \frac{\sqrt{3}}{4} = 0.433012701892219323381\ldots \neq \frac{2\sqrt{3}-1}{8} = 0.308012701892219323381\ldots. \tag{12}$$

6. ***Removable poles in $\partial_{q_i}\delta_{D,\mathrm{cl}}(3;q_i,\eta)$:*** *Because $\ell(p) = 3\sqrt{3}(4p^3 - p)$ satisfies $\ell_3 = \ell(p_{3,a}) = \ell(p_{3,b})$, the quotient*

$$\frac{\ell_3 - \ell(q_i)}{p_{3,a/b} - q_i} = 3\sqrt{3}\Big(4(p_{3,a/b}^2 + p_{3,a/b}q_i + q_i^2) - 1\Big) \tag{13}$$

is a polynomial with zero pole residue at $q_i = p_{3,a/b}$, refuting the simple-pole argument in Proposition 4.3 of `defq_charges_fusion.tex`.

2 Renormalised one-loop boundary theory and boundary renormalisation conditions

2.1 Exact real elementary closed forms for the defect Casimir charges

On any linearly stable defect vacuum $c = \eta\sigma_{\mathrm{vac}} \le -1$ ($\eta q_D \ge \frac{1}{2\sqrt{3}}$), there are no growing modes ($S > 1$), and the one-loop vacuum expectation values of the quadratic boundary momentum $P_{D,2}$ and quadratic energy $E_{D,2}$ are given by integrating the symmetric spectral traces over $q \in (0,\infty)$. By changing variables to $u = \sqrt{1+q^2} \in [1,\infty)$ (where $q\,\mathrm{d}q = u\,\mathrm{d}u$), the square root in $\omega(q) = qu$ and in $\theta'_D(\pm q;s,\eta) = -2\sqrt{3}\eta\frac{\sigma_{\mathrm{vac}}u\pm1}{u(4u^2\pm2\sigma_{\mathrm{vac}}u+\sigma_{\mathrm{vac}}^2-3)}$ rationalises completely **(c)** (`phase2/scripts/defb_boundary_renorm.py`).

Theorem 2.1 (Exact elementary closed forms for one-loop defect Casimir charges, **(c)**, `phase2/scripts/defb_boundary_renorm.py`)**.** *Let $\sigma = \sigma_{\mathrm{vac}} = -2\sqrt{3}q_D \in \mathbb{R}\setminus\{\pm1\}$ and $D_\pm(u) := 4u^2 \pm 2\sigma u + \sigma^2 - 3$. Then:*

1. *In the variable $u = \sqrt{1+q^2} \in [1,\infty)$ (or compact variable $z = 1/u \in (0,1]$), the UV-finite Casimir momentum $\Delta P_D^{\mathrm{Cas}}(\sigma,\eta) := \int_0^\infty (\langle P_{D,2}\rangle_q + \langle P_{D,2}\rangle_{-q})\,\mathrm{d}q$ and the UV-subtracted local Casimir energy $\mathcal{E}_D^{\mathrm{sub}}(\sigma,\eta) := \int_0^\infty \big[\langle E_{D,2}\rangle_q + \langle E_{D,2}\rangle_{-q} + \frac{\sqrt{3}\eta\sigma}{4\pi}\frac{q}{\sqrt{1+q^2}}\big]\,\mathrm{d}q$ reduce to the exact*

rational integrals

$$\Delta P_D^{\mathrm{Cas}}(\sigma,\eta) = \frac{2\sqrt{3}\,\eta}{\pi}\int_1^\infty \frac{(\sigma^2-2)u^2+\frac{1}{2}(3-\sigma^2)}{D_+(u)\,D_-(u)}\,\mathrm{d}u, \tag{14}$$

$$\mathcal{E}_D^{\mathrm{sub}}(\sigma,\eta) = \frac{\sqrt{3}\,\eta\,\sigma}{4\pi}\int_1^\infty \frac{-4u^2+(\sigma^2-3)^2}{D_+(u)\,D_-(u)}\,\mathrm{d}u. \tag{15}$$

2. *Evaluating (14)–(15) in closed form gives for $|\sigma|<2$:*

$$\Delta P_D^{\mathrm{Cas}}(\sigma,\eta) = \frac{\eta}{8\pi}\left[\sqrt{3}\,\sigma\ln\left|\frac{\sigma+1}{\sigma-1}\right| - \sqrt{4-\sigma^2}\,\ln\left|\frac{\sqrt{3}+\sqrt{4-\sigma^2}}{\sqrt{3}-\sqrt{4-\sigma^2}}\right|\right], \tag{16}$$

$$\Delta E_D^{\mathrm{ren}}(\sigma,\eta) = \frac{\eta}{16\pi}\left[-4\sqrt{3}\,\sigma+\sqrt{3}(2-\sigma^2)\ln\left|\frac{\sigma+1}{\sigma-1}\right| - \sigma\sqrt{4-\sigma^2}\,\ln\left|\frac{\sqrt{3}+\sqrt{4-\sigma^2}}{\sqrt{3}-\sqrt{4-\sigma^2}}\right|\right], \tag{17}$$

$$\mathcal{E}_D^{\mathrm{sub}}(\sigma,\eta) = \Delta E_D^{\mathrm{ren}}(\sigma,\eta) + \frac{\sqrt{3}\,\eta\,\sigma}{4\pi} = \frac{\eta}{16\pi}\left[\sqrt{3}(2-\sigma^2)\ln\left|\frac{\sigma+1}{\sigma-1}\right| - \sigma\sqrt{4-\sigma^2}\,\ln\left|\frac{\sqrt{3}+\sqrt{4-\sigma^2}}{\sqrt{3}-\sqrt{4-\sigma^2}}\right|\right], \tag{18}$$

and for $|\sigma|\geq 2$ (by smooth analytic continuation $\sqrt{4-\sigma^2}\,\operatorname{artanh}\frac{\sqrt{4-\sigma^2}}{\sqrt{3}} \mapsto -\sqrt{\sigma^2-4}\,\arctan\frac{\sqrt{\sigma^2-4}}{\sqrt{3}}$):

$$\Delta P_D^{\mathrm{Cas}}(\sigma,\eta) = \frac{\eta}{8\pi}\left[\sqrt{3}\,\sigma\ln\left|\frac{\sigma+1}{\sigma-1}\right| + 2\sqrt{\sigma^2-4}\,\arctan\frac{\sqrt{\sigma^2-4}}{\sqrt{3}}\right], \tag{19}$$

$$\Delta E_D^{\mathrm{ren}}(\sigma,\eta) = \frac{\eta}{16\pi}\left[-4\sqrt{3}\,\sigma+\sqrt{3}(2-\sigma^2)\ln\left|\frac{\sigma+1}{\sigma-1}\right| + 2\sigma\sqrt{\sigma^2-4}\,\arctan\frac{\sqrt{\sigma^2-4}}{\sqrt{3}}\right]. \tag{20}$$

At $\sigma=-2$ ($q_D=\frac{1}{\sqrt{3}}$), these reduce to the exact special values $\Delta P_D^{\mathrm{Cas}}(-2,\eta) = \frac{\sqrt{3}\ln 3}{4\pi}\eta$, $\Delta E_D^{\mathrm{ren}}(-2,\eta) = \frac{\sqrt{3}(4+\ln 3)}{8\pi}\eta$, and $\mathcal{E}_D^{\mathrm{sub}}(-2,\eta) = \frac{\sqrt{3}\ln 3}{8\pi}\eta$. Across $\sigma\in\{-1.2,-1.5,-\sqrt{3},-2.0,-2.5\}$, (16)–(20) agree with the two-root dipole representation to 1.36×10^{-71} and with compact-interval quadrature of (14)–(15) to 4.81×10^{-72}.

Table 1: One-loop renormalised defect Casimir charges on linearly stable vacua ($\eta=+1$, $c=\sigma_{\mathrm{vac}}<-1$) from Theorem 2.1, verified to 70-digit working precision (`phase2/scripts/defb_boundary_renorm.py`).

$\sigma_{\mathrm{vac}}=-2\sqrt{3}q_D$	$\Delta E_D^{\mathrm{ren}}(\sigma_{\mathrm{vac}},+1)$	$\mathcal{E}_D^{\mathrm{sub}}(\sigma_{\mathrm{vac}},+1)$	$\Delta P_D^{\mathrm{Cas}}(\sigma_{\mathrm{vac}},+1)$
-1.2	$+0.24243415056776965958$	$+0.07703548194123204473$	-0.00720648262722269233
-1.5	$+0.29997546003179577445$	$+0.09322712424862375589$	$+0.06055750652521987455$
$-\sqrt{3}$	$+0.32949203246641368672$	$+0.09075961782857068306$	$+0.10480017956974568334$
-2.0	$+0.35137653516192050810$	$+0.07571208745102448334$	$+0.15142417490204896669$
-2.5	$+0.36217135204446018208$	$+0.01759079240584015113$	$+0.23117575764210336842$

2.2 Envelope theorem on static vacua and boundary renormalisation conditions

At cutoff Λ ($T:=\sqrt{1+\Lambda^2}$), the equal-mode-count Casimir energy $E_D^{\mathrm{mode}}(\sigma_{\mathrm{vac}},\eta;\Lambda) = -\frac{\sqrt{3}\eta\sigma_{\mathrm{vac}}}{2\pi}(T-1) + \Delta E_D^{\mathrm{ren}}(\sigma_{\mathrm{vac}},\eta) + O(\Lambda^{-1})$ has a linear UV divergence proportional to $\sigma_{\mathrm{vac}} = s+\sqrt{3}\eta M_D$. Adding the unique UV boundary counterterm

$$\delta E_D^{\mathrm{ct,UV}}(\sigma;\Lambda) = +\frac{\sqrt{3}\,\eta(T-1)}{2\pi}\,\sigma = \frac{\sqrt{3}\,\eta(T-1)}{2\pi}\,s + \frac{3(T-1)}{2\pi}\left(\varphi_R(0^+)-\varphi_L(0^-)\right) \tag{21}$$

(or half this coefficient in the direct local density scheme $E_D^{\mathrm{dens}}(\Lambda)$) cancels the linear UV divergence and completes the open-half-line bulk normal-ordering tadpole $\frac{3(T-1)}{2\pi}\int_{x\neq 0}\varphi_x\,\mathrm{d}x = \frac{3(T-1)}{2\pi}(M_{\mathrm{tot}}-M_D)$ into the topological charge $\frac{3(T-1)}{2\pi}M_{\mathrm{tot}}$ **(c)**.

For the remaining finite $O(\hbar)$ boundary counterterms within the off-shell integrable CZ parameter family

$$(s,\,\rho_0,\,\tau_0,\,h_0)\;\longmapsto\;(s+\hbar\,\delta s_1,\;\rho_0+\hbar\,\delta\rho_{0,1},\;\tau_0+\hbar\,\delta\tau_{0,1},\;h_0+\hbar\,\delta h_{0,1}), \tag{22}$$

together with the bulk vacuum-elasticity counterterm $\Delta\mathcal{H}_{\mathrm{bulk}}=\hbar\lambda_g u^2$ of `phase2/onebody_scheme.tex` **(c)**, we establish the following exact variational formulas (`phase2/scripts/defb_boundary_renorm.py`, Sections 2–4):

Proposition 2.2 (Envelope theorem on static defect vacua and uniqueness of boundary renormalisation conditions, **(c)**, `phase2/scripts/defb_boundary_renorm.py`)**.** *Let $\sigma_i=-2\sqrt{3}q_i$ ($i=1,2,3$) be the three static vacuum roots of $\rho(\sigma_i)=\rho_0+\frac{\sqrt{3}\eta}{2}(\sigma_i-\sigma_i^3/3)=0$ for $s=-2\sqrt{3}q_D$.*

1. ***Envelope theorem on static vacua:*** *Because $\tau'(\sigma_i)=-\sigma_i\rho(\sigma_i)=0$, $h'(\sigma_i)=-\rho(\sigma_i)=0$, and $\rho(s)=0$, both the induced $O(\hbar)$ shifts $\delta\sigma_i$ of the static vacuum roots and the lower-limit shift δs_1 in $\tau(\sigma)=\tau_0-\int_s^\sigma z\rho(z)\,\mathrm{d}z$ and $h(\sigma)=h_0-\int_s^\sigma\rho(z)\,\mathrm{d}z$ contribute **zero** at $O(\hbar)$ to the static vacuum charges $E_D^{\mathrm{vac}}(q_i)=-\tau(\sigma_i)/9$ and $P_D^{\mathrm{vac}}(q_i)=h(\sigma_i)/9$. Reducing modulo the KP cubic $\sigma_i^3-3\sigma_i=s^3-3s$, the exact $O(\hbar)$ counterterm shifts of the static defect vacuum charges are*

$$\boxed{\delta E_D^{\mathrm{vac}}(q_i)=-\frac{1}{9}\delta\tau_{0,1}+\frac{2}{3}\delta\rho_{0,1}(q_i^2-q_D^2),\qquad \delta P_D^{\mathrm{vac}}(q_i)=\frac{1}{9}\delta h_{0,1}+\frac{2\sqrt{3}}{9}\delta\rho_{0,1}(q_i-q_D),} \tag{23}$$

   while the static root locations shift by $\delta q_i=\frac{\eta\,\delta\rho_{0,1}}{3(1-12q_i^2)}$.

2. ***Natural boundary renormalisation conditions and uniqueness:*** *The four finite boundary counterterm parameters $(\delta s_1,\delta\rho_{0,1},\delta\tau_{0,1},\delta h_{0,1})$ are uniquely fixed to $(0,0,0,0)$ by three physical boundary renormalisation conditions:*

   (a) Preservation of the $w_\varphi=0$ reference vacuum: *Requiring that $\varphi_L(0^-)-\varphi_R(0^+)=0$ ($\sigma=s_{\mathrm{ren}}$) remain a static vacuum solution of $\rho(\sigma)=0$ at $O(\hbar)$ uniquely fixes $\delta\rho_{0,1}=0$.*

   (b) Reference ground-state normalisation: *Requiring that the additive action constants at the reference vacuum $\sigma=s_{\mathrm{ren}}$ vanish ($\tau(s_{\mathrm{ren}})=h(s_{\mathrm{ren}})=0$, so the quantum ground-state charges are given purely by the spectral Casimir integrals $\Delta E_D^{\mathrm{ren}},\Delta P_D^{\mathrm{Cas}}$) uniquely fixes $\delta\tau_{0,1}=0$ and $\delta h_{0,1}=0$. (Alternative ground-state normalisation conventions shift all three vacua by the same constant and cancel in all vacuum jumps $Q_D(q_{i+1})-Q_D(q_i)$.)*

   (c) Zero-momentum transmission slope (scattering length): *At $q=0$, the phonon transmission amplitude (7) is topologically pinned to $z_{\mathrm{ph}}(0;s,\eta)=+1$ for every non-cusp vacuum $s\neq-1$, so the value at $q=0$ cannot fix δs_1. However, the zero-momentum phase derivative (defect scattering length)*

$$a_D(s,\eta):=\left.\frac{\mathrm{d}\theta_D^{\mathrm{phys}}(q;s,\eta)}{\mathrm{d}q}\right|_{q=0^+}=-\frac{2\sqrt{3}\,\eta}{s+1} \tag{24}$$

   is a strictly monotone one-to-one function of s on the stable half-line $\eta s\le-1$. Defining the renormalised defect parameter s_{ren} by $a_D^{\mathrm{ren}}(s_{\mathrm{ren}},\eta)=-\frac{2\sqrt{3}\eta}{s_{\mathrm{ren}}+1}$ (or by the physical soliton absorption root $q_D^{\mathrm{ren}}=-s_{\mathrm{ren}}/(2\sqrt{3})$) uniquely fixes $\delta s_1=0$.

3. ***Interaction with the bulk vacuum-elasticity counterterm*** $\Delta\mathcal{H}_{\rm bulk} = \hbar\lambda_g u^2$***:*** *On any spatially constant defect vacuum ($u_L = u_R = 0$), $\int_{x\neq 0} u^2\,\mathrm{d}x = 0$, so λ_g does not shift the defect Casimir charges. On a bulk soliton $S_i = (\alpha_i, \epsilon_i)$, adding $\Delta\mathcal{H}_{\rm bulk} = \hbar\lambda_g u^2$ leaves $\Delta P_{\rm mode}(S_i)$ unchanged, shifts $\Delta E_1^{\rm ren}(S_i)$ by $+\frac{2}{3}\lambda_g k_i^3$, and shifts the canonical pair by*

$$\delta_{\lambda_g}\Delta P_{\rm can}(S_i) = 2\lambda_g v_i k_i, \qquad \delta_{\lambda_g}\Delta E_{\rm can}(S_i) = \frac{2}{3}\lambda_g k_i(3 - 2k_i^2), \tag{25}$$

preserving $\mathrm{d}\Delta E_{\rm can} = v\,\mathrm{d}\Delta P_{\rm can}$ and the fixed-P energy shift $+\frac{2}{3}\lambda_g k_i^3$. Around any KP triangle $\{q_1, q_2, q_3\}$ with $\rho_i = \eta\,\mathrm{sgn}(q_i - q_{i+1})$, the cubic identity

$$\sum_{i=1}^{3}\rho_i\, k_{S_i}^3 = \eta\sum_{i=1}^{3}(q_i - q_{i+1})^3 = 3\eta\,(q_1 - q_2)(q_2 - q_3)(q_3 - q_1) \neq 0 \tag{26}$$

shows that $\lambda_g = 0$ (the continuum zero-sound-shift scheme) is the unique *value of λ_g for which the mode-sum soliton energies balance the analytically continued defect Casimir jumps across a KP triangle.*

Remark 2.3 (Scope of the counterterm classification, **(d)**). Following `phase2/audit_defq.tex` (Section 6), Proposition 2.2 classifies all integrable counterterms within the classical CZ functional family (s, ρ_0, τ_0, h_0) plus the bulk total-derivative and u^2 counterterms **(c)**. Whether higher-derivative quantum boundary operators outside the classical CZ ansatz could exist with anomalous quantum conservation laws is an open question beyond one-loop semiclassics **(d)**.

3 Recomputed $O(\hbar^0)$ transmission phase and absorption between stable vacua

3.1 Recomputed $O(\hbar^0)$ soliton–defect transmission phase on stable vacua

On any linearly stable defect ground state $q_{\rm st}$ ($c = \eta\sigma_{\rm st} = -2\sqrt{3}\eta q_{\rm st} \leq -1$), the renormalised boundary conditions $(\delta s_1, \delta\rho_{0,1}, \delta\tau_{0,1}, \delta h_{0,1}) = (0, 0, 0, 0)$ mean that no finite boundary counterterm shifts the classical transmission action $\frac{1}{\hbar}\delta_{D,\rm cl}(\alpha, \epsilon; q_{\rm st}, \eta)$ at fixed bare parameters. Expressing the one-loop transmission phase at fixed physical soliton momentum $P^{\rm phys} = P_{\rm cl}(\alpha, \epsilon) + \hbar\,\Delta P(\alpha, \epsilon)$ therefore gives **(c)** (`phase2/scripts/defb_boundary_renorm.py`):

$$\delta_D^{\rm trans,ren}(P^{\rm phys}; q_{\rm st}, \eta) = I_{SD}(\alpha, \epsilon; q_{\rm st}, \eta) - \Delta x_D(\alpha, \epsilon; q_{\rm st}, \eta)\,\Delta P(\alpha, \epsilon), \tag{27}$$

where $\Delta x_D(\alpha, \epsilon; q_{\rm st}, \eta) = -k^{-1}\ln|z_t(\alpha, \epsilon; q_{\rm st}, \eta)|^\epsilon$ and $I_{SD}(\alpha, \epsilon; q_{\rm st}, \eta) = -\int_{-\infty}^{\infty}\frac{\mathrm{d}q}{4\pi}\,\theta'_{\rm sol}(q; \alpha)\,\theta_D^{\rm asympt}(q; s_{\rm st}, \eta)$ with $s_{\rm st} = -2\sqrt{3}q_{\rm st}$ (Table 2). Under the bulk vacuum-elasticity counterterm $\Delta\mathcal{H}_{\rm bulk} = \hbar\lambda_g u^2$, $\delta_D^{\rm trans,ren}$ in the mode-sum scheme ($\Delta P = \Delta P_{\rm mode}$) is completely invariant, while in the canonical scheme ($\Delta P = \Delta P_{\rm can}$) it shifts by $-2\lambda_g v k\,\Delta x_D = +2\lambda_g v\ln|z_t|^\epsilon$ **(c)**.

Table 2: Recomputed $O(\hbar^0)$ soliton–defect transmission phase (27) for a regular test soliton ($\alpha_3 = 0.35$, $\epsilon_3 = +1$) across linearly stable defect vacua ($\eta = +1$, $q_{\rm st} \geq \frac{1}{2}$, $c = s_{\rm st} \leq -\sqrt{3}$) at $\delta s_1 = \lambda_g = 0$ (`phase2/scripts/defb_boundary_renorm.py`).

$q_{\rm st}$	$s_{\rm st} = -2\sqrt{3}q_{\rm st}$	$I_{SD}(3; q_{\rm st}, +1)$	$\delta_D^{\rm trans,ren}$ $(\Delta P_{\rm mode})$	$\delta_D^{\rm trans,ren}$ $(\Delta P_{\rm can})$
0.50	$-\sqrt{3} \approx -1.73205081$	-1.3321099117235175816	-2.8344357811760363607	-3.7897181712546800328
0.52	-1.8013328398716323853	-1.3633755953997418592	-2.6721305722452243578	-3.5043272399121963223

3.2 Why stable-vacuum absorption balance reduces to closed KP cycles

Proposition 3.1 (Monotonicity proof that no single absorption edge connects two stable vacua, **(c)**, `phase2/scripts/defb_boundary_renorm.py`)**.** *For a fixed defect orientation $\eta \in \{+1, -1\}$, linear stability ($c = -2\sqrt{3}\eta q \leq -1$) requires $\eta q \geq \frac{1}{2\sqrt{3}}$. On the entire stable half-line $|q| > \frac{1}{2\sqrt{3}}$, the KP cubic polynomial $f(q) = 4q^3 - q$ has strictly positive derivative*

$$f'(q) = 12q^2 - 1 > 0 \qquad \forall\, |q| > \frac{1}{2\sqrt{3}}. \tag{28}$$

Consequently, $f(q)$ is strictly injective on the stable domain $\eta q \geq \frac{1}{2\sqrt{3}}$, so ***no two distinct roots of*** $4q^3 - q = 4q_D^3 - q_D$ ***can both lie in the linearly stable regime for the same orientation*** *η.*

Proposition 3.1 has a direct consequence for testing absorption balance **between stable vacua only**:

1. At fixed orientation η, any elementary single-soliton transition $D_\eta(q_i) + S_i \to D_\eta(q_{i+1})$ has at least one linearly unstable endpoint ($c \in (-1, 2)$).

2. The *only* absorption/emission processes whose external asymptotic defect states are linearly stable vacua at fixed η are **closed cycles returning to the unique stable vacuum** $q_{\rm st}$, namely the composite two-soliton fusion transition

$$D_\eta(q_{\rm st}) + S_1 + S_2 \;\longleftrightarrow\; D_\eta(q_{\rm st}) + \bar{S}_3 \tag{29}$$

   (or the three-soliton loop $D_\eta(q_{\rm st}) + S_1 + S_2 + S_3 \leftrightarrow D_\eta(q_{\rm st})$). Because the initial and final defect state in (29) is the *identical* stable vacuum $D_\eta(q_{\rm st})$, the defect vacuum charge $Q_D^{\rm quant}(q_{\rm st})$ cancels out completely between the initial and final states, without ever requiring a quantum charge assignment on the two unstable vertices!

3. Therefore, one-loop energy–momentum conservation in the stable-to-stable composite process (29) at unshifted classical absorption rapidities reduces *purely* to the vanishing of the directed KP-triangle holonomy $(\mathcal{C}_E, \mathcal{C}_P)$ of the three bulk solitons S_1, S_2, S_3 **(c)**.

4 Universal symbolic and interval proof of the KP-triangle holonomy theorem

4.1 Elementary closed-form reduction of $(\Delta E_{\rm can}, \Delta P_{\rm can})$ and $(\mathcal{C}_E(\vartheta), \mathcal{C}_P(\vartheta))$

We first reduce the bulk soliton co-moving effective Lagrangian $L_1(\alpha) = v\Delta P_{\rm mode}(\alpha, \epsilon) - \Delta E_1^{\rm ren}(\alpha)$ and canonical collective-coordinate charges $(\Delta E_{\rm can}, \Delta P_{\rm can})$ to elementary functions of $k = \sin\alpha$ and $v = \epsilon\cos\alpha$ **(c)** (`phase2/scripts/defb_boundary_holonomy.py`, Section 1):

Proposition 4.1 (Elementary closed forms for L_1, $\Delta E_{\rm can}$, and $\Delta P_{\rm can}$, **(c)**, `phase2/scripts/defb_boundary_holonomy.py`)**.** *For every $\alpha \in (0, \frac{\pi}{2}) \setminus \{\frac{\pi}{3}\}$ and $\epsilon \in \{+1, -1\}$, with $k = \sin\alpha$ and $v = \epsilon\cos\alpha$:*

1. *The tangent ratio in $\Delta P_{\rm can}(\alpha, \epsilon)$ obeys the exact trigonometric identity*

$$\frac{\tan\left(\frac{\alpha+2\pi/3}{2}\right)}{\tan\left(\frac{\alpha-2\pi/3}{2}\right)} \equiv \frac{2\sin\alpha + \sqrt{3}}{2\sin\alpha - \sqrt{3}} = \frac{2k + \sqrt{3}}{2k - \sqrt{3}}. \tag{30}$$

2. *In $L_1(\alpha) = v\Delta P_{\rm mode} - \Delta E_1^{\rm ren}$, the third root $\phi_3 = \alpha$ drops out identically because $\cos\alpha - \cos\phi_3 = 0$, while the remaining two roots $\phi_{1,2} = \alpha \pm \frac{2\pi}{3}$ satisfy $\sin(\alpha \pm \frac{2\pi}{3})\big(\cos\alpha - \cos(\alpha \pm$*

$\frac{2\pi}{3})) \equiv \pm\frac{\sqrt{3}}{4}(3-4k^2)$. *Consequently,* L_1, ΔP_{can}, *and* ΔE_{can} *have the exact closed forms (verified to* 5.07×10^{-70}*):*

$$L_1(k) = \frac{3k}{2\pi} - \frac{\sqrt{3}}{8\pi}(3-4k^2)\ln\left|\frac{2k+\sqrt{3}}{2k-\sqrt{3}}\right|, \tag{31}$$

$$\Delta P_{\mathrm{can}}(k,v) = -\frac{\sqrt{3}}{\pi}\,v\,\ln\left|\frac{2k+\sqrt{3}}{2k-\sqrt{3}}\right|, \tag{32}$$

$$\Delta E_{\mathrm{can}}(k) = -\frac{3k}{2\pi} - \frac{\sqrt{3}}{8\pi}(5-4k^2)\ln\left|\frac{2k+\sqrt{3}}{2k-\sqrt{3}}\right|. \tag{33}$$

Every nondegenerate KP triple $\{q_1,q_2,q_3\}$ ($q_1+q_2+q_3=0$, $q_1q_2+q_2q_3+q_3q_1=-\frac{1}{4}$) is parametrised up to cyclic relabelling and orientation reversal by the fundamental domain $\vartheta\in(0,\frac{\pi}{3})$ via **(c)** (`phase2/scripts/defb_boundary_holonomy.py`, Section 2):

$$q_j(\vartheta) = -\frac{1}{\sqrt{3}}\cos\left(\vartheta+\frac{2\pi j}{3}\right)\quad(j=1,2,3),\qquad 4q_j(\vartheta)^3 - q_j(\vartheta) = -\frac{\cos(3\vartheta)}{3\sqrt{3}}. \tag{34}$$

On the entire open interval $\vartheta\in(0,\frac{\pi}{3})$, the three roots maintain the strict ordering

$$q_3(\vartheta) < -\frac{1}{2\sqrt{3}} < q_2(\vartheta) < +\frac{1}{2\sqrt{3}} < q_1(\vartheta), \tag{35}$$

so the edge signs $\rho_i = \eta\,\mathrm{sgn}(q_i-q_{i+1})$ are constant across $(0,\frac{\pi}{3})$: $\rho_1=+\eta$, $\rho_2=+\eta$, $\rho_3=-\eta$, and the three edge solitons $S_i=\{q_i,q_{i+1}\}$ have exact parameters

$$\begin{aligned}
S_1:&\quad k_1(\vartheta)=\sin\vartheta, && v_1(\vartheta)=+\cos\vartheta, && \rho_1=+\eta,\\
S_2:&\quad k_2(\vartheta)=\sin\left(\frac{\pi}{3}-\vartheta\right), && v_2(\vartheta)=-\cos\left(\frac{\pi}{3}-\vartheta\right), && \rho_2=+\eta,\\
S_3:&\quad k_3(\vartheta)=\sin\left(\vartheta+\frac{\pi}{3}\right)=k_1+k_2, && v_3(\vartheta)=-\cos\left(\vartheta+\frac{\pi}{3}\right), && \rho_3=-\eta.
\end{aligned} \tag{36}$$

4.2 Universal proof across all KP triangles

Theorem 4.2 (Universal canonical-charge holonomy theorem for all KP triangles, **(c)**, `phase2/scripts/defb_boundary_holonomy.py`)**.** *For every* $\vartheta\in(0,\frac{\pi}{3})$ *and* $\eta\in\{+1,-1\}$*, the canonical-charge KP-triangle holonomies* $\mathcal{C}_E(\vartheta) := \sum_{i=1}^3 \rho_i\Delta E_{\mathrm{can}}(S_i)$ *and* $\mathcal{C}_P(\vartheta) := \sum_{i=1}^3 \rho_i\Delta P_{\mathrm{can}}(S_i)$ *satisfy:*

1. ***Exact closed forms:*** *Because* $\sum_{i=1}^3\rho_ik_i = \eta(k_1+k_2-k_3)\equiv 0$*, the* $-\frac{3k_i}{2\pi}$ *linear tadpole in (33) cancels identically around every KP triangle, giving (verified against vertex sums to* 5.46×10^{-70}*):*

$$\mathcal{C}_E(\vartheta) = -\frac{\sqrt{3}\,\eta}{8\pi}\left[(5-4k_1^2)\ln\frac{\sqrt{3}+2k_1}{\sqrt{3}-2k_1} + (5-4k_2^2)\ln\frac{\sqrt{3}+2k_2}{\sqrt{3}-2k_2} - (5-4k_3^2)\ln\frac{2k_3+\sqrt{3}}{2k_3-\sqrt{3}}\right], \tag{37}$$

$$\mathcal{C}_P(\vartheta) = -\frac{\sqrt{3}\,\eta}{\pi}\left[v_1\ln\frac{\sqrt{3}+2k_1}{\sqrt{3}-2k_1} + v_2\ln\frac{\sqrt{3}+2k_2}{\sqrt{3}-2k_2} - v_3\ln\frac{2k_3+\sqrt{3}}{2k_3-\sqrt{3}}\right], \tag{38}$$

where all three logarithm arguments are strictly greater than 1 *on* $\vartheta\in(0,\frac{\pi}{3})$*.*

2. ***Reflection symmetry and exact values at the symmetric triple*** $\vartheta=\frac{\pi}{6}$ ***(***$\{q_1,q_2,q_3\}=\{\frac{1}{2},0,-\frac{1}{2}\}$***):*** *Under reflection* $\vartheta\mapsto\frac{\pi}{3}-\vartheta$ *across the midpoint* $\vartheta=\frac{\pi}{6}$*,* $\mathcal{C}_E(\vartheta)$ *is* even *and*

$\mathcal{C}_P(\vartheta)$ *is* odd, *with exact midpoint values*

$$\mathcal{C}_E\left(\frac{\pi}{6}\right) = -\frac{3\sqrt{3}}{4\pi}\ln(2+\sqrt{3})\,\eta = -0.54455770697142409838087 14\ldots\,\eta \neq 0, \tag{39}$$

$$\mathcal{C}_P\left(\frac{\pi}{6}\right) = 0, \qquad \mathcal{C}'_P\left(\frac{\pi}{6}\right) = \frac{3\sqrt{3}\ln(2+\sqrt{3})-9}{\pi}\,\eta = -0.68655814776841965031642 23\ldots\,\eta \neq 0. \tag{40}$$

3. ***Exact endpoint asymptotics and cancellation identity:*** *For all* $\vartheta \in (0, \frac{\pi}{3})$, *the difference of the cusp logarithms* $L_2 := \ln\frac{\sqrt{3}+2k_2}{\sqrt{3}-2k_2}$ *and* $L_3 := \ln\frac{2k_3+\sqrt{3}}{2k_3-\sqrt{3}}$ *obeys the exact singularity-free identity*

$$L_3(\vartheta) - L_2(\vartheta) \equiv \ln\left(\frac{2k_3+\sqrt{3}}{\sqrt{3}+2k_2}\cdot\frac{1+\sqrt{3}\tan(\vartheta/2)}{1-\sqrt{3}\tan(\vartheta/2)}\right), \tag{41}$$

and as $\vartheta \to 0^+$ *(and symmetrically as* $\vartheta \to (\frac{\pi}{3})^-$*),*

$$\mathcal{C}_E(\vartheta) = -\frac{3\eta}{2\pi}\,\vartheta\left(\ln\frac{2\sqrt{3}}{\vartheta}+1\right) + O(\vartheta^2\ln\vartheta), \qquad \mathcal{C}_P(\vartheta) = +\frac{3\eta}{\pi}\,\vartheta\left(\ln\frac{2\sqrt{3}}{\vartheta}-2\right) + O(\vartheta^2\ln\vartheta). \tag{42}$$

4. ***Universal non-vanishing on*** $(0, \frac{\pi}{3})$***:*** *Combining analytical bounds on* $(0, 10^{-4}]$ *(where* $-\eta\mathcal{C}_E(\vartheta)/\vartheta > 5.46$ *and* $\eta\mathcal{C}_P(\vartheta)/\vartheta > 8.07$*), rigorous* `mpmath.iv` *interval evaluation of* (37)–(41) *across* 2000 *subintervals of* $[10^{-4}, \frac{\pi}{6} - 10^{-3}]$ *(certifying* $\eta\mathcal{C}_E(\vartheta) \le -4.02\times10^{-4} < 0$ *and* $\eta\mathcal{C}_P(\vartheta) \ge +1.17\times10^{-4} > 0$*), and rigorous interval bounds on* $[\frac{\pi}{6} - 10^{-3}, \frac{\pi}{6}]$ *(certifying* $\eta\mathcal{C}_E(\vartheta) < -0.54 < 0$ *and* $\eta\mathcal{C}'_P(\vartheta) < -0.68 < 0$ *with* $\mathcal{C}_P(\frac{\pi}{6}) = 0$*):*

$$\boxed{\eta\,\mathcal{C}_E(\vartheta) < 0 \quad \forall\,\vartheta \in \left(0, \frac{\pi}{3}\right), \qquad \mathcal{C}_P(\vartheta) = 0 \iff \vartheta = \frac{\pi}{6}, \qquad (\mathcal{C}_E(\vartheta), \mathcal{C}_P(\vartheta)) \neq (0,0) \quad \forall\,\vartheta \in \left(0, \frac{\pi}{3}\right).} \tag{43}$$

5. ***Exact invariance of*** $\mathcal{C}_P(\vartheta)$ ***under the bulk vacuum-elasticity counterterm*** λ_g***:*** *Because* $\rho_i v_i k_i = \sqrt{3}\eta(q_i^2 - q_{i+1}^2)$ *telescopes to zero around every triangle (*$\sum_{i=1}^3 \rho_i v_i k_i \equiv 0$*), the canonical momentum holonomy is completely independent of* λ_g*:*

$$\mathcal{C}_P^{\lambda_g}(\vartheta) \equiv \mathcal{C}_P(\vartheta), \qquad \mathcal{C}_E^{\lambda_g}(\vartheta) = \mathcal{C}_E(\vartheta) + \eta\,\lambda_g\sin(3\vartheta) \qquad \forall\,\lambda_g \in \mathbb{R}. \tag{44}$$

In particular, for every $\vartheta \neq \frac{\pi}{6}$, $\mathcal{C}_P^{\lambda_g}(\vartheta) \neq 0$ *for all* $\lambda_g \in \mathbb{R}$, *and at* $\vartheta = \frac{\pi}{6}$ *the vector* $(\mathcal{C}_E^{\lambda_g}(\frac{\pi}{6}), \mathcal{C}_P^{\lambda_g}(\frac{\pi}{6}))$ *is non-zero in both the continuum zero-sound-shift scheme (*$\lambda_g = 0$*) and the Toda-matched scheme (*$\lambda_g = \frac{9}{4\pi} \approx 0.716197 \neq \frac{3\sqrt{3}}{4\pi}\ln(2+\sqrt{3}) \approx 0.544558$*).*

4.3 Universal proof for the absorbed-soliton momentum remainder and vertex-shift derivative

Finally, we upgrade Proposition 4.3 of `phase2/defq_charges_fusion.tex` to a universal proof across all nondegenerate KP triangles while replacing its refuted simple-pole argument (`audit_defq.tex` Correction (vi)) by the exact pole-free derivative identity **(c)** (`phase2/scripts/defb_boundary_holonomy.py`, Section 4):

Proposition 4.3 (Universal non-vanishing of $\mathcal{R}_{\rm abs}$ and exact vertex-derivative identity, **(c)**, `phase2/scripts/defb_boundary_holonomy.py`)**.** *For any nondegenerate KP triangle* $\vartheta \in (0, \frac{\pi}{3})$ *and test soliton* $3 = (\alpha_3, \epsilon_3)$*, let* $\mathcal{R}_{\rm abs}(\alpha_3, \epsilon_3; \vartheta) := \sum_{i=1}^3 \rho_{{\rm tree},i}\,\Delta X_{S_i}(3, S_i)\,\Delta P(S_i) = \sum_{i=1}^3 (w_{i-1} - w_i)\ln|z_t(3; q_i, \eta)|$ *where* $w_i := \Delta P(S_i)/k_{S_i}$*.*

1. ***Universal strict sign separation*** $w_1(\vartheta) < 0 < w_2(\vartheta)$***:*** *On every nondegenerate KP triangle* $\vartheta \in (0, \frac{\pi}{3})$*, edge* S_1 *has* $\alpha_1 = \vartheta \in (0, \frac{\pi}{3})$ *and* $\epsilon_1 = +1$ *(*$v_1 > 0$*), while edge* S_2 *has* $\alpha_2 = \frac{\pi}{3} - \vartheta \in (0, \frac{\pi}{3})$ *and* $\epsilon_2 = -1$ *(*$v_2 < 0$*). Because* $\Delta P_{\mathrm{can}}(\alpha, +1) < 0$ *and* $\Delta P_{\mathrm{mode}}(\alpha, +1)/\sin\alpha \le -\frac{\ln 3}{2\pi} < -0.17 < 0$ *for all* $\alpha \in (0, \frac{\pi}{3})$ *(certified analytically and across* 500 *interval subdivisions), every* $\vartheta \in (0, \frac{\pi}{3})$ *satisfies*

$$\boxed{w_1(\vartheta) < 0 < w_2(\vartheta) \quad \Longrightarrow \quad w_1(\vartheta) - w_2(\vartheta) < 0 \qquad \forall\, \vartheta \in \left(0, \frac{\pi}{3}\right),} \tag{45}$$

for both $\Delta P = \Delta P_{\mathrm{can}}$ *and* $\Delta P = \Delta P_{\mathrm{mode}}$ *(as well as for any* $\lambda_g \ge 0$*, since* $\delta_{\lambda_g} w_i = 2\lambda_g v_i$ *is only negative if* $\lambda_g < 0$*). Because* $\ln|z_t(3; q_2, \eta)| = \eta \ln\left|\frac{p_{3,b} - q_2}{p_{3,a} - q_2}\right|$ *has logarithmic branch points at* $p_{3,a/b} = q_2(\vartheta)$ *that are disjoint from* $q_1(\vartheta)$ *and* $q_3(\vartheta)$*,* $\mathcal{R}_{\mathrm{abs}}(\alpha_3, \epsilon_3; \vartheta)$ *is a non-zero function of* α_3 *on* ***every*** *nondegenerate KP triangle.*

2. ***Exact pole-free formula for*** $\partial_{q_i}\delta_{D,\mathrm{cl}}(3; q_i, \eta)$ ***and telescoping of vertex shifts:*** *On the soliton root locus* $\ell_3 = \ell(p_{3,a}) = \ell(p_{3,b})$*, the apparent poles* $\frac{\ell_3 - \ell(q_i)}{p_{3,a/b} - q_i}$ *in the derivative of* $\delta_{D,\mathrm{cl}}(3; q_i, \eta) = \frac{2}{9}(\ell_3 - \ell(q_i))\epsilon_3 \ln|z_t(3; q_i, \eta)| - \frac{2}{3}(\Pi_3 M_D(q_i) - \Pi_D(q_i) M_3)$ *cancel identically, yielding the exact smooth identity*

$$\boxed{\frac{\partial \delta_{D,\mathrm{cl}}(3; q_i, \eta)}{\partial q_i} = -\frac{2\sqrt{3}}{3}(12 q_i^2 - 1)\epsilon_3 \ln\left|z_t(3; q_i, \eta)\right| + 4\epsilon_3 \eta\, k_3 (v_3 + 2\sqrt{3}\, q_i).} \tag{46}$$

Any single-valued one-loop shift $q_i \mapsto q_i + \hbar\, \delta q_i$ *of the defect vertices telescopes identically to zero around the closed triangle (*$\sum_{i=1}^{3} \left[\partial_{q_{i+1}} \delta_{D,\mathrm{cl}}(3; q_{i+1}, \eta)\delta q_{i+1} - \partial_{q_i}\delta_{D,\mathrm{cl}}(3; q_i, \eta)\delta q_i\right] \equiv 0$*) and therefore cannot cancel* $\mathcal{R}_{\mathrm{abs}}$ *around a closed KP triangle.*

Remark 4.4 (Qualified physical scope of the triangle obstructions, **(d)**). Theorem 4.2 and Proposition 4.3 settle the mathematical question left open by `phase2/audit_defq.tex`: there is **no counterexample KP triangle** on which $(\mathcal{C}_E, \mathcal{C}_P)$ or $\mathcal{R}_{\mathrm{abs}}$ vanishes at the unshifted classical absorption rapidities **(c)**. As stressed in `phase2/audit_defq.tex`, the physical interpretation remains qualified **(d)**: if a future non-perturbative quantum bound-state or resonance construction establishes the $O(\hbar)$ absorption-rapidity shift $a_{1,i} = (\Delta P_{\mathrm{mode}}(S_i) - \Delta P_{\mathrm{can}}(S_i))/P'_{\mathrm{cl}}(S_i)$ and modifies the fixed-momentum fusion prescription across unstable defect resonances, closure could be restored via those shifted kinematics **(d)**.

5 Summary of claims, verification scripts, and failures recorded

5.1 Summary table of claims and checks

Claim / Formula	Label	Script	Precision / Tolerance
Off-shell conservation of $(E, P, \Pi, M)_{\mathrm{tot}}$ for arbitrary (s, ρ_0, τ_0, h_0) and $\eta = \pm 1$	**(c)**	`phase2/scripts/defb_boundary_renorm.py`	Exact symbolic (`SymPy`)
True $z_t = 0$ absorption conservation vs old script's $s = +1.5$, $z_{\mathrm{wrong}} = \sqrt{21} - 3$	**(c)**	`phase2/scripts/defb_boundary_renorm.py`	2.26×10^{-72} (70 dps `mpmath`)
Quadratic energy trace plus sign (10), $\langle J_{\mathrm{bulk},2}\rangle_q = 0$, and $z_{\mathrm{ph}}(\pm\infty) = \mathrm{e}^{-2\pi i\eta/3}$	**(c)**	`phase2/scripts/defb_boundary_renorm.py`	Exact symbolic (`SymPy`)

Claim / Formula	Label	Script	Precision / Tolerance
Branch-switch growth rate $g_1(\sqrt{3}) = \sqrt{3}/4 \neq (2\sqrt{3}-1)/8$ and removable pole (13)	**(c)**	`phase2/scripts/defb_boundary_renorm.py`	Exact symbolic (`SymPy`)
Envelope theorem on static vacua: $\delta\sigma_i, \delta s_1$ give zero $O(\hbar)$ shift of $(E_D^{\mathrm{vac}}, P_D^{\mathrm{vac}})$ (23)	**(c)**	`phase2/scripts/defb_boundary_renorm.py`	Exact polynomial reduction (`SymPy`)
Zero-momentum pinning $z_{\mathrm{ph}}(0) = 1$ and scattering length $a_D(s,\eta) = -2\sqrt{3}\eta/(s+1)$ fixing $\delta s_1 = 0$	**(c)**	`phase2/scripts/defb_boundary_renorm.py`	Exact symbolic (`SymPy`)
Exact elementary closed forms (16)–(20) for $(\Delta E_D^{\mathrm{ren}}, \mathcal{E}_D^{\mathrm{sub}}, \Delta P_D^{\mathrm{Cas}})$	**(c)**	`phase2/scripts/defb_boundary_renorm.py`	1.36×10^{-71} vs dipole; 4.81×10^{-72} vs quadrature
Bulk elasticity shift (25) and KP triangle cube sum (26) selecting $\lambda_g = 0$ for mode-sum balance	**(c)**	`phase2/scripts/defb_boundary_renorm.py`	Exact symbolic (`SymPy`)
Monotonicity $f'(q) = 12q^2 - 1 > 0$ on $\lvert q\rvert > \frac{1}{2\sqrt{3}}$ and recomputed $O(\hbar^0)$ transmission phase (27)	**(c)**	`phase2/scripts/defb_boundary_renorm.py`	Exact symbolic & 70 dps `mpmath`
Elementary closed forms (31)–(33) for $L_1(k)$, $\Delta P_{\mathrm{can}}(k,v)$, $\Delta E_{\mathrm{can}}(k)$	**(c)**	`phase2/scripts/defb_boundary_holonomy.py`	Exact symbolic & 5.07×10^{-70}
Universal closed-form holonomies $(\mathcal{C}_E(\vartheta), \mathcal{C}_P(\vartheta))$ (37)–(38) and midpoint/endpoint values (39)–(42)	**(c)**	`phase2/scripts/defb_boundary_holonomy.py`	Exact symbolic & 5.46×10^{-70}
Universal proof that $\eta\mathcal{C}_E(\vartheta) < 0$ on $(0, \frac{\pi}{3})$ and $\mathcal{C}_P(\vartheta) = 0 \iff \vartheta = \frac{\pi}{6}$, plus λ_g-invariance of $\mathcal{C}_P$	**(c)**	`phase2/scripts/defb_boundary_holonomy.py`	Analytical bounds + 2000 `mpmath.iv` intervals
Universal proof that $w_1(\vartheta) < 0 < w_2(\vartheta)$ in $\mathcal{R}_{\mathrm{abs}}$ and exact vertex derivative (46)	**(c)**	`phase2/scripts/defb_boundary_holonomy.py`	Exact symbolic + 500 `mpmath.iv` intervals
Exhaustive non-CZ quantum boundary counterterms and resonance continuation of unstable vacua	**(d)**	—	Qualified open physical questions

5.2 Failures recorded

In accordance with `AGENTS.md`, we record both the six audit-flagged errors from `phase2/defq_charges_fusion.` that are corrected here and the intermediate script failures encountered and fixed during the development of `phase2/scripts/defb_boundary_renorm.py` and `phase2/scripts/defb_boundary_holonomy.py`:

1. **Hand-typed decimal literal for $z_{\mathrm{wrong}} = \sqrt{21} - 3$ (`phase2/scripts/defb_boundary_renorm.py`):** In the initial draft of `phase2/scripts/defb_boundary_renorm.py`, z_{wrong} at $q_1 = \sqrt{3}/4$, $\eta = -1$ was compared to a decimal string typed beyond the 21st digit, failing at 4.33×10^{-22} against the tolerance 10^{-55}. Derived the exact algebraic value $z_{\mathrm{wrong}} =$

$\frac{12}{3+\sqrt{21}} = \sqrt{21} - 3$ and verified agreement to $< 10^{-70}$.

2. **Derivative part of $\langle E_{D,2}\rangle_q$ copied from $\langle P_{D,2}\rangle_q$ (`phase2/scripts/defb_boundary_renorm.py`):** In `defq_oneloop_renorm.py` (line 293), the contraction of $a_x^2 + a_x b_x + b_x^2$ in $E_{D,2}$ was mistyped as $q^2(1 + 2\,\mathrm{Re}\, z)$ (the integrand of $\frac{1}{2}(a_x^2 + b_x^2) + 2a_x b_x$ from $P_{D,2}$) instead of $q^2(2 + \mathrm{Re}\, z)$, causing $\langle E_{D,2}\rangle_{\mathrm{sym}} - \langle K\rangle_{\mathrm{sym}} \neq 0$. Replacing $1 + 2\,\mathrm{Re}\, z$ by the correct $2 + \mathrm{Re}\, z$ makes $\langle E_{D,2}\rangle_{\mathrm{sym}} \equiv \langle K\rangle_{\mathrm{sym}} \equiv +\frac{\omega}{4\pi}(\theta'_D(q) + \theta'_D(-q))$ hold identically.

3. **Polynomial reduction modulo the KP cubic in the envelope-theorem check (`phase2/scripts/defb_boundary_renorm.py`):** Using `.subs(sig_sym**3, ...)` in `SymPy` did not reduce the quartic power σ_i^4 in $\partial_\epsilon \tau(\sigma_i + \epsilon\delta\sigma_i)$, leaving an unreduced remainder in $\partial_{\delta\sigma_i}\delta E_D^{\mathrm{vac}}$. Replacing `.subs` by exact polynomial remainder reduction `sp.rem(..., sig_sym**3 - 3*sig_sym - (s**3 - 3*s), sig_sym)` reduced the derivative identically to zero.

4. **Sign of $\cos\psi_{1,2}$ and analytic continuation across $|\sigma| = 2$ in $\Delta E_D^{\mathrm{ren}}$ (`phase2/scripts/defb_boundary_renorm.py`):** An initial sign swap between $\cos(\frac{2\pi}{3} - \gamma) = \frac{-\sigma + \sqrt{3(4-\sigma^2)}}{4}$ and $\cos(-\frac{2\pi}{3} - \gamma)$ gave -1 instead of $\frac{2-\sigma^2}{2}$ in the coefficient of $\ln\left|\frac{\sigma+1}{\sigma-1}\right|$ in $\Delta E_D^{\mathrm{ren}}$ (failing against the dipole sum at $\sigma = -1.2$ by 0.15839), and using $\arctan\frac{\sqrt{3}}{\sqrt{\sigma^2-4}}$ instead of $\arctan\frac{\sqrt{\sigma^2-4}}{\sqrt{3}}$ for $|\sigma| > 2$ introduced a branch offset of $\frac{15}{64} = 0.234375$ at $\sigma = -2.5$. Fixing both signs and using the smooth continuation $\sqrt{4-\sigma^2}\,\mathrm{artanh}\,\frac{\sqrt{4-\sigma^2}}{\sqrt{3}} \mapsto -\sqrt{\sigma^2-4}\arctan\frac{\sqrt{\sigma^2-4}}{\sqrt{3}}$ together with the exact cancelled rational integrand (15) on $z = 1/u \in [0,1]$ gave 70-digit agreement (4.81×10^{-72}).

5. **Interval dependency near the cusp logarithms L_2, L_3 as $\vartheta \to 0^+$ (`phase2/scripts/defb_boundary_holonomy.py`):** Evaluating $(5 - 4k_2^2)L_2 - (5 - 4k_3^2)L_3$ in `mpmath.iv` as two separate interval logarithms on $[10^{-4}, 3.6 \times 10^{-4}]$ suffered from interval dependency (± 0.177). Factoring $L_3 - L_2$ via the exact singularity-free identity (41) (and similarly factoring $\frac{2\cos\alpha - 1}{\sqrt{3} - 2\sin\alpha} = \tan(\frac{\alpha}{2} + \frac{\pi}{6})$ in w_{mode} on $[\frac{\pi}{6}, \frac{\pi}{3}]$) eliminated the endpoint dependency completely.

6. **Factor of 2 in the polynomial part of $\partial_{q_i}\delta_{D,\mathrm{cl}}(3; q_i, \eta)$ (`phase2/scripts/defb_boundary_holonomy.py`):** In the initial target expression for $\partial_{q_i}\delta_{D,\mathrm{cl}}(3; q_i, \eta)$, substituting $M_3 = 2k_3$ into $4\sqrt{3}M_3 q_i = 8\sqrt{3}k_3 q_i$ was written as $4\epsilon_3\eta k_3(v_3 + \sqrt{3}q_i)$ instead of $4\epsilon_3\eta k_3(v_3 + 2\sqrt{3}q_i) = 4\epsilon_3\eta k_3(v_3 - s_i)$. Correcting the coefficient to $2\sqrt{3}q_i$ gave exact symbolic identity.

References

[1] E. Corrigan and C. Zambon, *Integrable defects in the Boussinesq equations*, J. Phys. A: Math. Theor. **56** (2023) 345202, arXiv:2306.03772 (full text read in `literature/`).

[2] L. D. Faddeev and V. E. Korepin, *Quantum theory of solitons*, Phys. Rep. **42** (1978) 1–87, doi:10.1016/0370-1573(78)90058-3 (full text read in `literature/`).